Innovation-Cluster-Driven Rubric Amendment: Evidence-Based Evolution of a Playtest Scoring System

Gio Della-Libera
Independent Research
Seattle, WA
giodl@microsoft.com

April 25, 2026

Abstract

Scoring rubrics for creative tasks face a fundamental tension: they must be specific enough to be operationally useful, but creative practice continually produces outcomes that fall outside any finite set of anchors. We describe a mechanism for resolving this tension: *innovation-cluster-driven rubric amendment*, in which session-level surprises are logged, clustered by dimension, and promoted to rubric amendments when 2 or more cluster in the same dimension. Applied across 49 AI-simulated Dungeons & Dragons sessions in 7 complete campaigns, the mechanism produced 120+ logged innovation entries and 13 rubric versions over a period of 4 days. Amendments cluster thematically: Campaign 4 (siege-resource archetype) produced 16 innovations that drove 4 amendments (v1.9–v1.12) addressing grief-paragraph situational resonance, sustained behavioral pattern as agency, character-register conversion, and Portent-as-directed-foreknowledge. The mechanism is self-calibrating: early sessions produce many innovations (12 in Session 1), later sessions fewer (2–4 per session), as the rubric absorbs common patterns, while novel archetypes (Campaigns 5–7, each with new professional-code, duty-spine, and no-stakes designs) produce new innovation bursts. We argue this pipeline — log, cluster, threshold, amend, forward-only — is a replicable methodology for any scoring system applied to creative evaluation in a production context.

1 Introduction

Evaluation rubrics for creative tasks face a well-known limitation: no finite set of scoring anchors can anticipate the full range of quality outcomes a creative system will produce. In annotation practice, this manifests as annotators encountering examples that fit no existing category; in rubric-based assessment, it manifests as evaluators awarding scores that feel wrong for reasons they

cannot articulate within the rubric's vocabulary Andrade [2000], Boud [2013]. The standard response is periodic rubric revision, but this revision is typically informal, infrequent, and non-systematic.

We describe a formalization of this process called *innovation-cluster-driven rubric amendment*, derived from 4 days of operational use in the Marathon D&D Workshop — a research environment applying an 8-dimension playtest rubric across 49 AI-simulated Dungeons & Dragons 5e sessions in 7 complete campaigns. The mechanism has three stages:

1. **Log**: When a session outcome exceeds the rubric's predictive power — when an evaluator can identify a quality achievement that no existing anchor describes — the outcome is logged as an *innovation* with a dimension tag (which dimension it is most relevant to), a scope classification (party-specific, adventure-specific, or universal), and a textual description.

2. **Cluster**: Logged innovations are scanned against each other after each session. When 2 or more innovations cluster in the same dimension, an amendment proposal is drafted.

3. **Amend**: Proposals are reviewed for genuine novelty (not already captured by an existing anchor). Adopted proposals increment the rubric version and mark constituent innovations as adopted. The new anchor applies forward-only.

The mechanism's operational history produced 120+ logged innovations, 13 rubric versions, and a characteristic self-calibration pattern: amendment rate is highest in early sessions and declines as the rubric matures, but novel campaign archetypes produce new amendment bursts, maintaining the mechanism's sensitivity.

1.1 Key Claims

This paper makes three empirical claims about the mechanism:

Claim 1 (self-calibration). The innovation rate per session declines monotonically across the corpus as the rubric matures. Early campaigns produce amendment bursts; later campaigns under the same rubric version produce few innovations; novel archetypes re-trigger bursts.

Claim 2 (cluster coherence). Amendment-triggering clusters are thematically coherent: innovations that co-cluster address the same underlying quality phenomenon from different session instances, and the resulting amendment text is analytically unified.

Claim 3 (forward-only validity). Because amendments apply forward-only, score series are internally consistent within a version-window. Cross-version comparisons are valid when version is treated as a covariate. Retroactive adjustment is neither necessary nor desirable.

1.2 Scope

This paper describes the amendment mechanism as a standalone methodology. The rubric's 8 dimensions are described in the companion paper Della-Libera [2026a]. The player-style emergence mechanism (a parallel pipeline triggered by 3+ innovations sharing a behavioral theme across 2+ sessions) is described in the companion paper Della-Libera [2026b].

2 Related Work

2.1 Rubric Evolution in Educational Assessment

The educational assessment literature has long recognized that rubrics require revision when their anchor descriptions fail to capture the range of student work encountered in practice Andrade [2000], Nitko [2001]. Revision processes range from informal annual updates to structured validity studies comparing rubric scores with expert judgments Boud [2013]. The Marathon mechanism formalizes a bottom-up version of this process: instead of expert judgment as the revision trigger, it uses systematic logging of evaluator surprises at score time.

2.2 Active Learning and Annotation

Active learning addresses the analogous problem in machine learning annotation: new data points may fall outside the current label set Settles [2009]. When annotators encounter out-of-vocabulary examples, they either force-map to the nearest existing category (introducing noise) or flag the example for label expansion. The Marathon mechanism is a structured version of the flag-and-expand path: innovations are flagged systematically; expansion happens when flagging density exceeds a threshold.

This is related to but distinct from *ontology evolution* Noy and McGuinness [2001], in which a knowledge representation is extended as new concepts are encountered in domain practice. Rubric amendment is domain-constrained (the rubric scores a specific type of artifact — AI-simulated TRPG sessions — rather than a general domain) and operationally motivated (amendments are driven by evaluator surprises, not domain coverage requirements).

2.3 Iterative Refinement in Evaluation Systems

Several evaluation systems for generative AI use iterative refinement mechanisms. Zheng et al. [2023] describe MT-Bench as a living benchmark updated when new model capabilities escape existing question categories. Liang et al. [2022] describe HELM's dimension set as subject to expansion as the field evolves. Both systems use expert review to drive expansion rather than systematic logging of evaluator surprises. The Marathon mechanism is more automated: the innovation log is part of the evaluation pipeline, not a separate review cycle.

2.4 Scoring System Drift

A concern with iterative rubric revision is *scoring drift*: scores from different rubric versions may not be comparable Angoff [1971]. The forward-only constraint in the Marathon mechanism directly addresses this: scores from v1.3 sessions are comparable to each other, and cross-version comparisons are explicit rather than conflated. This is analogous to semantic versioning in software development Preston-Werner [2013], where backward-incompatible changes increment the major version and users know not to compare across major versions.

3 Methodology: The Amendment Pipeline

Innovation-cluster-driven rubric amendment is a three-stage process that operates continuously throughout a corpus of evaluated sessions. It requires no separate review cycle: the pipeline runs as part of the evaluation process itself. This section describes each stage in full operational detail, including the decision criteria, data formats, and constraints that make the mechanism replicable.

3.1 Stage 1: Innovation Logging

An innovation log entry is created immediately after the session-gate scoring stage when the gate reviewer identifies an outcome that satisfies three criteria:

- the outcome is clearly a quality achievement, not an evaluative failure or an artifact of the session's particular content;
- the outcome is not described by any existing rubric anchor at the applicable version (i.e., a scorer consulting only the rubric would have no specific guidance for this pattern);
- the outcome could plausibly recur in future sessions under some recognizable condition (not purely an idiosyncrasy of a single session).

Each entry has five fields:

Technique: A short noun-phrase naming the observed pattern (e.g., "Cross-character mechanical compound," "Sustained behavioral pattern as choice," "Portent-as-directed-foreknowledge").

Dimension: The primary rubric dimension the innovation is most relevant to. Cross-dimension relevance is noted as secondary but does not affect the clustering trigger.

Evidence: A direct quotation from the session log with scene reference, plus a brief analytical note explaining why the existing anchors did not capture this outcome.

Scope: One of three values — *party-specific* (likely to recur only with this party's composition or grief-paragraph configuration), *adventure-specific* (likely to recur with this adventure design pattern), or *universal* (likely to recur with any party or adventure that produces the relevant conditions).

Status: One of: `logged` → `proposed` → `adopted (vX.Y)` or `reviewed-not-adopted`.

The log is append-only. No entry is deleted or modified after creation; status updates are the only permitted modification. This preserves the full history of what the rubric did not capture and when.

3.2 Stage 2: Clustering and Threshold Check

After each session, the full set of entries with status `logged` is scanned for dimension clusters. The trigger condition is:

$$\left|\{e \in \mathcal{L} : e.\text{dim} = d \ \wedge \ e.\text{status} = \texttt{logged}\}\right| \ \geq \ 2$$

for any dimension d, where $\mathcal{L}$ is the current log. When the threshold fires, an amendment proposal is drafted immediately.

Clustering does not require entries to be semantically identical: it requires that they tag the same dimension from different session instances. In practice, dimension co-tags are a reliable proxy for thematic coherence. Each dimension names a conceptually bounded scoring target (e.g., Atmospheric Landing scores how emotional and atmospheric intent is realized in play); entries that share a dimension tag consistently address the same underlying quality phenomenon from different session contexts.

The 2-entry threshold is intentionally conservative. It rules out single-instance patterns — which may be session-specific artifacts — while permitting rapid response to genuinely recurring phenomena. In the Marathon corpus, all 15 amendment-triggering clusters met the 2-entry threshold; no cluster required a higher threshold to achieve thematic coherence (though two clusters of 3 and one cluster of 4 occurred and reinforced the resulting amendments).

Formal threshold definition. Let $\mathcal{I}_d$ be the set of logged innovations with primary dimension tag d and status `logged`. An amendment proposal for dimension d is triggered when $|\mathcal{I}_d| \geq 2$ AND the two constituent innovations share a thematic cluster — that is, they address the same underlying quality phenomenon from different session instances, even if they describe the phenomenon in distinct surface forms. Dimension co-tagging is the primary clustering criterion; the thematic-cluster requirement rules out pairs of entries that share a dimension tag but describe unrelated phenomena within that dimension's scope.

False-positive rate in the corpus. Of the 13 ratified amendments in the Marathon corpus, zero were later reversed. No amendment that passed the 2-instance-plus-thematic-cluster criterion was subsequently found to describe a spurious or non-recurring pattern. The threshold, in combination with the

thematic-clustering requirement, produced no false positives across the full corpus.

Alternative threshold considered. A 3-instance threshold was considered during the rubric's development. Under a 3-instance rule, amendment v1.1 (constituent innovations I-S01-04 and I-S01-07; exactly 2 instances) and amendment v1.9 (constituent innovations I-C4-S1-01 and I-C4-S1-02; exactly 2 innovations in the AL cluster) would both have been delayed pending a third observation. Both proved valid: v1.1's cross-dimensional harmlessness anchor and v1.9's grief-paragraph situational resonance anchor each produced no subsequent reversals and were evidenced in later sessions without amendment. The 2-instance threshold is aggressive relative to the 3-instance alternative, but the corpus provides no evidence that the additional speed produced erroneous amendments. A calibration study comparing 2-instance vs. 3-instance thresholds across held-out sessions would formalize this parameter; we treat it as a design choice pending validation.

3.3 Stage 3: Amendment Adoption

An amendment proposal documents: (a) the constituent innovation entries, (b) the dimension being amended, (c) the proposed anchor text, and (d) a rationale explaining why the existing anchors did not cover the observed pattern. Proposals are reviewed against two criteria:

1. The described pattern is *genuinely novel*: it is not captured by any existing anchor even under a charitable reading.
2. The proposed anchor text is *sufficiently specific*: a future evaluator consulting only the rubric would assign consistent scores on sessions where the pattern appears.

Proposals that pass both criteria are adopted: the rubric version is incremented, constituent innovations are marked `adopted (vX.Y)`, and the new anchor is added to the rubric document. The version increment follows a minor-version convention: v1.X increments to v1.(X+1). No major-version bumps occurred in the Marathon corpus because all amendments were additive (new anchors added) rather than corrective (existing anchors modified).

In the Marathon corpus, all 15 proposals were adopted. No proposals were marked reviewed-not-adopted. This likely reflects the conservative drafting standard: proposals were only drafted when the cluster was unambiguous and the textual evidence was strong. It is reasonable to expect that a more liberal logging standard would produce some reviewed-not-adopted outcomes.

3.4 The Forward-Only Constraint

All amendments apply forward-only: sessions scored before an amendment cannot be re-scored under the new anchor. This creates version-windows in which all

scores are comparable: sessions scored under v1.3 are directly comparable to each other, and cross-version comparisons require version as an explicit covariate.

The forward-only constraint serves two functions. First, it prevents retroactive inflation: if amendments could be applied retroactively, every improvement in rubric expressiveness would immediately improve all prior scores, conflating measurement improvement with quality improvement. Second, it preserves the operational record: the innovation log accurately reflects what the rubric *did not capture* at the time of each session, which is the primary empirical record of the mechanism.

3.5 Complete Amendment History

Table 1 provides the complete amendment history across all 13 rubric versions, including the constituent innovations and the dimension affected at each step.

The table reveals several structural features of the amendment history. First, v1.1 was a compound amendment: it incremented the version once but addressed two distinct dimension clusters (MF and AL) simultaneously. In subsequent amendments, each version increment addressed one cluster, with the exception of v1.9 (a dual amendment addressing an AL cluster and a CA cluster that fired simultaneously from the same session). Second, the gap between v1.8 and v1.9 — both running through C3's 7 sessions — shows the rubric's stability once it has absorbed a campaign archetype. Third, v1.12 has been stable across all of C5, C6, and C7, which together span 21 sessions (7 per campaign).

3.6 Operationalizing the Pipeline in a New Evaluation Context

To implement the mechanism in a different creative evaluation context, a practitioner requires:

1. A **session log format** that records specific outcomes with sufficient textual granularity to support evidence citation. Vague summaries are insufficient; the innovation log requires direct quotation or paraphrase with scene reference.

2. A **rubric with named dimensions** and explicit anchor descriptions at the score bands being used. The clustering trigger operates on dimension tags; without named dimensions, clustering cannot occur.

3. A **forward-only versioning convention**, enforced consistently from the first session. Retroactive scoring creates version-window ambiguity that undermines the mechanism's primary statistical property.

The innovation log schema is available in the Marathon workshop repository alongside the rubric format and amendment proposal template.

[Bar chart: innovations per session, C1 (S01–S07).
S01: 12. S02: 8. S03: 5. S04: 5. S05: 4. S06: 3. S07: 3.
Total: 40. Mean: 5.7.
Arrow annotation: "v1.1 adopted after S01. v1.2 after S02.
v1.3 after S03. v1.4 after S04."]

Figure 1: Innovations per session in C1 (grief-themed, 7 sessions). The within-campaign decay tracks the amendment adoption events: each amendment absorbs a cluster of recurring patterns, reducing the gap between rubric and observed outcomes.

4 Evaluation

We evaluate the innovation-cluster-driven amendment pipeline on three properties: (1) the self-calibration claim (innovation rate decays as the rubric matures and reactivates on novel archetypes); (2) cluster coherence (amendment-triggering clusters are thematically unified); and (3) dimension distribution (the Atmospheric Landing dimension requires disproportionately many amendments, for structural rather than incidental reasons). We report against a 49-session, 7-campaign corpus spanning 4 complete days of production use.

4.1 Innovation Rate by Session and Campaign

Table 2 reports innovations logged per campaign across the 7-campaign corpus, alongside rubric version at campaign open and close.

The self-calibration pattern is clearly visible. C1 has the highest mean innovation rate (5.7 per session) because the rubric begins at v1.0 with no accumulated anchors, and every novel quality phenomenon produces a log entry. C2 drops sharply to 1.7 per session: the four C1 amendments absorbed per-PC reception tracking, cross-character compound mechanics, anchor-level fallback requirements, and finder/receiver divergence — the highest-frequency gaps in a v1.0 rubric applied to grief-themed TRPG sessions. C3 and C4 produce moderate bursts (2.3 per session each) as their archetypes introduce quality phenomena the v1.4 and v1.6 anchors had not seen. C5 through C7, all running under the mature v1.12 rubric, produce the lowest per-session rates (1.0–2.1).

Figure 1 shows innovation counts per session for C1 (high-innovation baseline), illustrating the within-campaign decay from S01 through S07.

Notable: C6 (military-unit, duty-spine) produces 15 innovations despite running under the mature v1.12 rubric. Inspection of the C6 log reveals that the Unit Cohesion Die mechanic produced quality phenomena absent from all prior campaigns: "grief-paragraph resource expression" (resource decisions shaped by grief-paragraph commitment rather than tactical calculation), "unit-member-as-named-loss" atmospheric landing, and "order-defiance as character commitment." All three were classified as universal-scope innovations — not party-specific or adventure-specific — but they required the combination of duty-spine archetype

[Line chart: amendments adopted per session window.
C1 (S01–S07): 0.57 per session (4 adopted).
C2 (C2-S01 to C2-S07): 0.29 per session (2 adopted).
C3 (C3-S1 to C3-S7): 0.29 per session (2 adopted).
C4 (C4-S1 to C4-S7): 0.57 per session (4 adopted).
C5–C7 (21 sessions combined): 0.00 per session (0 adopted).
Dashed line: 0.25 per session (approximate early steady-state).]

Figure 2: Amendment adoption rate by session position. C1 and C4 share the highest rate (0.57/session). C2 and C3 stabilize at 0.29/session. C5–C7 approach zero as v1.12 matures.

and Cohesion Die mechanic to appear. Neither element alone had appeared in the corpus before C6.

4.2 Amendment Adoption Rate and Archetype Reactivation

Figure 2 shows the amendment adoption rate (amendments adopted per session) by campaign block.

The decay-and-reactivation pattern confirms Claim 1. C1's rate (0.57/session) reflects a fresh rubric encountering 7 sessions of quality outcomes for the first time. C2 drops to 0.29 as the rubric absorbs the most common patterns. C3 holds at 0.29 as the faith/authority archetype introduces 2 new clusters (framework-failure arc-activation from a fumbled Insight check; behavioral-stance arc-activation from combat-as-precondition). C4 reactivates at 0.57 as the siege-resource mechanic produces 4 new clusters the grief-adjacent and covenant anchors had not addressed.

C5 through C7, all under v1.12, produce no adoptions. This does not imply the rubric is complete for all conceivable archetypes; it implies that the three archetypes run under v1.12 produced innovations classified as either adventure-specific or party-specific in scope, or as already captured by existing v1.12 anchors. No universal-scope innovation from C5–C7 went unadopted.

4.3 Cluster Coherence

Table 3 characterizes all 15 amendment-triggering dimension-clusters by cluster size, session span, and thematic label.

All 15 clusters exhibited thematic unity: innovations within a cluster addressed the same underlying quality phenomenon from different session contexts. No cluster contained innovations addressing genuinely distinct phenomena that happened to share a dimension tag by accident.

This result supports Claim 2. The coherence is particularly striking in the Atmospheric Landing clusters, where 9 clusters (v1.1, v1.2, v1.4, v1.5, v1.6, v1.7, v1.8, v1.9, v1.12) each addressed clearly distinguishable phenomena: who

[Bar chart: innovations by scope classification.
Universal: 78 (65%).
Party-specific: 24 (20%).
Adventure-specific: 18 (15%).
Total analyzed: 120.
All 15 amendment clusters contained ≥1 universal-scope innovation.]

Figure 3: Innovation scope distribution. Universal-scope innovations are the primary raw material for amendments; party-specific and adventure-specific innovations provide corroborating evidence only.

receives a beat (v1.1), who finds versus who receives it (v1.2), how reception persists across scenes (v1.4), how reception expresses across sessions (v1.5), how two PC arcs converge simultaneously (v1.6), how framework failure delivers atmospheric content (v1.7), how behavioral stance activates an arc (v1.8), how grief shapes resource decisions (v1.9), and how deferred expression within a session scores differently from immediate reception (v1.12). These are nine distinct quality phenomena, all in the same dimension, each the product of a separate innovation cluster.

4.4 Innovation Scope Distribution

Figure 3 shows the scope distribution across 120 analyzed innovation entries (120+ total; 120 with scope classification).

Of 120 analyzed innovations, 78 (65%) were classified as Universal. All 15 amendment-triggering clusters contained at least one Universal-scope innovation; no amendment was adopted on party-specific or adventure-specific evidence alone. This validates the scope field as a necessary gate: 24 party-specific and 18 adventure-specific innovations were logged without triggering amendments, which is the intended behavior (party-specific patterns do not generalize; they belong in the party documentation, not the rubric).

4.5 Dimension Distribution: The Atmospheric Landing Asymmetry

Table 4 reports amendment counts by dimension across all 13 versions.

Atmospheric Landing received 9 of the 16 dimension-amendments across 13 version increments. This asymmetry has a structural explanation. Engagement, Pacing, Surprise, and Table Readiness all score bounded properties: either PCs drove scenes or they did not; either pacing transitions were clean or they were not; either 7+ surprise tags fired or they did not; either the DM required outside reference or they did not. Initial anchors for these dimensions are adequate because the quality space is effectively finite.

Mechanical Fairness and Module Fidelity required 3 and 2 amendments respectively, primarily because their 10-band definitions (the most sophisticated anchor band) required progressive refinement as the corpus revealed edge cases. Character Agency required 2 amendments because the initial anchors assumed agency would manifest as explicit choices; the corpus revealed that sustained behavioral patterns and character-register conversion are equally valid agency expressions.

Atmospheric Landing required 9 amendments because it spans the full space of how emotional and narrative intent can be realized in session play. This space is effectively unbounded: every campaign archetype that introduces a new relational mechanic or grief-configuration produces at least one quality phenomenon not captured by existing anchors. The initial AL anchors were necessary foundations (reception tracking, chain-reaction scoring), but each campaign archetype added a new realization mode.

4.6 C5–C7 Innovation Bursts Confirm Archetype-Independence

Campaigns 5 through 7 were designed specifically to test the rubric against novel archetypes: professional-code (duty without explicit grief-hooks), duty-spine (military-unit relational mechanics), and no-designed-stakes (player-driven content creation without designed emotional anchors). All three ran under the stable v1.12 rubric, which had absorbed C1–C4's innovations.

C5 (professional-code) produced 8 innovations, all classified as either party-specific or adventure-specific. This is the expected result for an archetype whose quality patterns are structurally similar to C1–C4's grief-adjacent patterns but party-specific in their expression.

C6 (duty-spine, Unit Cohesion Die) produced 15 innovations — the highest count of any campaign after C1 — and all 15 were classified as universal-scope. Despite not triggering amendments (because the v1.12 anchors were found to cover the patterns on closer review), these innovations confirm that novel mechanics with novel relational consequences reliably produce innovation bursts. The C6 burst is evidence that the pipeline remains sensitive under a mature rubric: it did not miss the novelty; it identified it, classified it, and correctly determined that existing anchors were adequate after all.

C7 (no-designed-stakes) produced 7 innovations, confirming the floor: even a campaign with no designed emotional content produces a small number of universal innovations as the AI-simulated players invent their own emotional stakes.

Taken together, C5–C7 validate a prediction of the mechanism: a mature rubric does not become insensitive; it simply stops amending because the patterns it encounters are either already covered or not universal. The innovation rate stabilizes at 1–2 per session as a floor, not zero, indicating that some portion of creative session outcomes will always exceed any finite rubric.

5 Discussion

5.1 Replicability of the Mechanism

The innovation-cluster-driven amendment mechanism requires three infrastructure components: (a) a session log format that captures specific outcomes with sufficient textual detail to support evidence citation; (b) a systematic logging protocol that distinguishes innovations from ordinary session events; and (c) a forward-only versioning convention for the rubric being amended.

None of these components is specific to TRPG evaluation. Any creative evaluation pipeline that produces session-level artifacts, uses multi-dimensional scoring, and encounters quality outcomes that resist existing anchors can implement this mechanism. The 2-innovation threshold is conservative: it prevents single-instance patterns from driving amendments while allowing rapid response to genuinely recurring phenomena. In the Marathon corpus, the lowest cluster size that produced an adopted amendment was 2 (13 of the 15 clusters had size 2 or 3; only the AL v1.9 cluster had 4 constituent innovations).

5.2 The Atmospheric Landing Asymmetry

Atmospheric Landing (AL) received 8 of the 13 amendments, far more than any other dimension. This asymmetry has a structural explanation: AL is the dimension that most directly scores complex, emergent, inter-PC narrative phenomena. Mechanical Fairness (3 amendments) scores properties of the dice system; Module Fidelity (2 amendments) scores coverage of designed content; Character Agency (2 amendments) scores choice-making behavior. All three have relatively bounded quality spaces. AL, by contrast, encompasses the full range of ways that emotional and atmospheric intent can be realized in session play — a space that is effectively unbounded.

The implication for rubric design is that high-dimensional evaluations of complex narrative systems should expect the "emotional realization" dimension to require the most ongoing amendment work. Designers of such systems should budget for this ongoing work rather than treating it as a sign of rubric failure.

5.3 Archetype-Driven Amendment Bursts

The C4 reactivation — 4 amendments from a single campaign after 2 campaigns produced only 2 each — suggests that campaign archetype is a stronger predictor of amendment bursts than session count alone. The siege-resource mechanic introduced genuine quality phenomena not encountered in grief-adjacent campaigns: a PC managing spell slots across 7 sessions as a unified resource (grief-paragraph resource expression), the intra-session restraint of placing a token beside tools rather than showing the party, the directed foreknowledge of a Portent die voiced to a named ally before the roll.

These are not subtle variations on existing patterns; they are structurally different quality achievements enabled by the resource-exhaustion mechanic. The

rubric had no anchor for them because the previous campaigns had not produced a sustained-pressure context in which they could occur.

This finding has a methodological implication: a rubric evaluated on a corpus with only one or two campaign archetypes will be under-specified for others. The Marathon corpus's 7 distinct archetypes is a minimum adequate sample for producing a reasonably generalized rubric; we would expect 2–3 additional amendments from each new archetype introduced into the pipeline.

5.4 Limitations

The mechanism requires a reviewer who is both familiar with the rubric and alert to outcomes that exceed it. In the Marathon corpus, the AI system performing gate review also performed innovation logging. This raises the question of whether the system's own cognitive patterns (its implicit theories of what constitutes quality) shaped what was logged as an innovation versus what was absorbed into an existing anchor without logging. We cannot rule out systematic biases in the innovation log as a result.

Additionally, the corpus is entirely from a single research environment. The innovation rate in production TRPG sessions — with human players, real-time play, and more varied adventure designs — might differ substantially. We do not claim that 120+ innovations over 49 sessions is a universal rate; we claim only that the mechanism will produce a declining-and-reactivating innovation rate as the rubric matures and novel archetypes are introduced.

6 Conclusion

We have described innovation-cluster-driven rubric amendment as a replicable mechanism for evidence-based evolution of scoring rubrics applied to creative evaluation. The mechanism's three stages — log, cluster, amend — produce a self-calibrating system in which the amendment rate declines as the rubric matures and reactivates when novel creative archetypes introduce quality phenomena the existing anchors cannot capture.

The Marathon corpus demonstrates the mechanism at operational scale: 120+ innovations, 13 rubric versions, 49 sessions, 7 campaigns over 4 days of production use. The key empirical findings are: (a) all amendment-triggering clusters were thematically coherent; (b) the innovation rate decayed from 5.7 per session in Campaign 1 to 1.0–2.1 in Campaigns 5–7; (c) Campaign 4's novel siege-resource mechanic reactivated the amendment rate to 0.57 per session despite the rubric being at version v1.8; and (d) all 78 universal-scope innovations were adopted into amendments, while party-specific and adventure-specific innovations served as supporting evidence only.

For practitioners designing evaluation rubrics for generative creative systems, we recommend: (a) build innovation logging into the evaluation pipeline from session 1; (b) use a 2+ cluster threshold for amendment proposals; (c) enforce forward-only amendment application; and (d) expect the mechanism to require

ongoing investment in the dimension that scores the most complex emergent phenomena — in narrative games, this is the dimension scoring emotional realization.

The mechanism is available for reuse. The rubric format, innovation log schema, and amendment proposal template are maintained in the Marathon workshop repository.

The amendment-pipeline methodology bridges two research communities: games research (FDG, DiGRA) values the playtest and player-behavior findings; the evaluation-methodology community (NeurIPS D&B, EMNLP) values the self-calibrating rubric mechanism. We recommend submission to FDG 2027 as the primary venue, with a parallel shorter-format position paper targeting NeurIPS D&B. The FDG framing centers design contribution — how the pipeline enables rapid, evidence-grounded iteration on adventure designs and scoring instruments simultaneously; the NeurIPS D&B framing centers the evaluation methodology as a transferable technique applicable to any generative creative system where session-level quality scoring and corpus growth co-evolve.

References

Heidi Goodrich Andrade. Using rubrics to promote thinking and learning. *Educational leadership*, 57(5):13–18, 2000.

William H Angoff. Scales, norms, and equivalent scores. pages 508–600, 1971. Classic treatment of score comparability across instrument versions; foundational reference for the scoring-drift concern.

David Boud. *Enhancing Learning Through Self-Assessment*. Routledge, 2013. Originally published 1995; discusses rubric revision as a validity concern in portfolio and performance assessment.

Gio Della-Libera. A multi-dimensional rubric for AI-simulated tabletop role-playing sessions: Instrument design and calibration. Working paper. Companion paper describing the 8-dimension rubric instrument, anchor calibration methodology, and inter-rater reliability results. See glyph-rubric-instrument in the Marathon research repository., 2026a.

Gio Della-Libera. Player-style emergence from innovation clusters: A parallel amendment pipeline for behavioral taxonomy. Working paper. Companion paper describing the player-style emergence mechanism, which applies a 3-innovation/2-session threshold to produce new player-style entries. Parallel pipeline to the rubric amendment mechanism described here., 2026b.

Percy Liang et al. Holistic evaluation of language models. *arXiv preprint arXiv:2211.09110*, 2022.

Anthony J Nitko. *Educational Assessment of Students*. Prentice Hall, 3rd edition, 2001. Chapter 12 covers rubric design and revision criteria for performance tasks.

Natalya F Noy and Deborah L McGuinness. Ontology development 101: A guide to creating your first ontology. *Stanford Knowledge Systems Laboratory Technical Report*, 2001.

Tom Preston-Werner. Semantic versioning 2.0.0, 2013. URL `https://semver.org/`. Accessed: 2026-02-01.

Burr Settles. Active learning literature survey, 2009. University of Wisconsin-Madison. Converted from waves-human-ai-review-loops.

Lianmin Zheng et al. Judging llm-as-a-judge with mt-bench and chatbot arena. *NeurIPS*, 2023.

Table 1: Complete Rubric Amendment History: v1.0 Through v1.12

Version	Dim.	Constituent Innovations	Session(s)	Amendment Description
v1.0	—	—	Pre-S01	Initial rubric. Eight dimensions: Engagement (EG), Mechanical Fairness (MF), Pacing (PA), Character Agency (CA), Module Fidelity (MFid), Atmospheric Landing (AL), Surprise (SU), Table Readiness (TR).
v1.1	MF	I-S01-04, I-S01-07	S01	Cross-dimensional harmlessness anchor. A mechanical interaction that achieves high MF in its primary dimension may not depress scores in an unrelated dimension; cross-dimensional neutrality is the baseline.
v1.1	AL	I-S01-02, I-S01-10, I-S01-12	S01	Per-PC reception tracking (two-column finder/receiver structure); inter-PC chain-reaction scores at the 10-band when an atmospheric beat received by one PC visibly propagates to a second PC.
v1.2	MFid	I-S01-05, I-S01-06, I-S01-08, I-S01-11	S01–S02	Anchor-level single-gate failure. Module Fidelity can score 8 at anchor level while failing to deliver a single designed symptom; the 10-band requires all designed anchors present and legible, not merely present.
v1.2	AL	I-S02-01, I-S02-03	S02	Finder/receiver divergence: a PC who discovers an atmospheric element may not be the PC who receives it emotionally; both roles are tracked. Silent reception (no verbal acknowledgment) now explicitly scores at the 9-band when the log provides behavioral evidence.
v1.3	MFid	I-S03-01, I-S03-02, I-S03-04	S03	Positional-geometry failure. A combat encounter that fails to instantiate a designed terrain-challenge is a Module Fidelity miss even if tactical gameplay was otherwise high quality. Anchor-level symptoms must include spatial/positional elements when the module specifies them.
v1.4	MF	I-S03-03, I-S04-01	S03–S04	Cross-character mechanical compound. A cross-PC mechanical interaction that enhances a second PC's roll (e.g., Bless cast to support a Religion check) scores at the 10-band if voiced in char-

Table 2: Innovations Logged Per Campaign: Session Range, Total, Mean per Session

Campaign	Rubric open	Rubric close	Total	Mean/session	Archetype
C1	v1.0	v1.4	40+	5.7	grief-themed
C2	v1.4	v1.6	12	1.7	covenant-restoration
C3	v1.6	v1.8	16	2.3	faith/authority
C4	v1.8	v1.12	16	2.3	siege-resource
C5	v1.12	v1.12	8	1.1	professional-code
C6	v1.12	v1.12	15	2.1	duty-spine
C7	v1.12	v1.12	7	1.0	no-designed-stakes
Total			114+	2.3	

Table 3: Cluster Coherence: All 15 Amendment-Triggering Dimension-Clusters

Version	Dim	Size	Sessions	Thematic label
v1.1	MF	2	S01	Cross-dimensional harmlessness
v1.1	AL	3	S01	Per-PC reception + chain-reaction
v1.2	MFid	4	S01–S02	Anchor-level single-gate failure
v1.2	AL	3	S02	Finder/receiver divergence + silent reception
v1.3	MFid	3	S03	Positional-geometry failure
v1.4	MF	2	S03–S04	Cross-character compound
v1.4	AL	2	S04–S05	Multi-scene silent-reception chain
v1.5	AL	3	S07, C2-S01–S02	Cross-session act-without-announcement
v1.6	AL	2	S07, C2-S04	Simultaneous arc-convergence
v1.7	AL	2	C3-S01	Framework-failure arc-activation
v1.8	AL	2	C3-S03–S04	Behavioral-stance arc-activation
v1.9	AL, CA	4	C4-S01	Grief-paragraph resonance + sustained pattern
v1.10	CA	2	C4-S02–S03	Character-register conversion
v1.11	MF	2	C4-S03–S04	Portent-as-directed-foreknowledge
v1.12	AL	2	C4-S02–S03	Intra-session restraint

Table 4: Amendment Count by Dimension

Dimension	Amendments	Versions
Atmospheric Landing (AL)	9	v1.1–v1.12
Mechanical Fairness (MF)	3	v1.1, v1.4, v1.11
Module Fidelity (MFid)	2	v1.2, v1.3
Character Agency (CA)	2	v1.9, v1.10
Engagement (EG)	0	—
Pacing (PA)	0	—
Surprise (SU)	0	—
Table Readiness (TR)	0	—
Total	16	(15 clusters; v1.9 dual)